

RFM ANALYSIS

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RFM is a customer segmentation technique based on three dimensions of customer purchasing behavior:

- **Recency — How recently did the customer purchase?**
- **Frequency — How often does the customer purchase?**
- **Monetary — How much does the customer spend?**

Objective

Identify different types of customers and design appropriate marketing strategies.

Transaction Data → RFM Metrics → RFM Score → Customer Segments → Marketing Action

Customer Segmentation Using RFM

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Common business segments include:

- Champions

Recent + frequent + high-value customers

- Loyal Customers

Frequently purchase and show consistent engagement

- Potential Loyalists

Recently active but have room to increase purchase frequency

- At Risk

Previously valuable/active customers who have not purchased recently

- Lost Customers

Low recent activity and low engagement/value



RFM to Marketing Action

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Segment	Possible Marketing Action
Champions	Rewards, exclusive benefits
Loyal Customers	Loyalty programs, personalized offers
Potential Loyalists	Cross-sell and repeat-purchase offers
At Risk	Re-engagement campaigns
Lost Customers	Win-back campaigns

Market Basket Analysis

Market Basket Analysis

Market Basket Analysis identifies products that tend to be purchased together.

Example-

A supermarket discovers:

Bread → Butter

A customer purchasing Bread may also purchase Butter.

Applications

- Cross-selling
- Product recommendations
- Combo offers
- Promotions
- Store layout
- E-commerce recommendations



Association Rules

An association rule represents a relationship between items.

General structure

$A \rightarrow B$

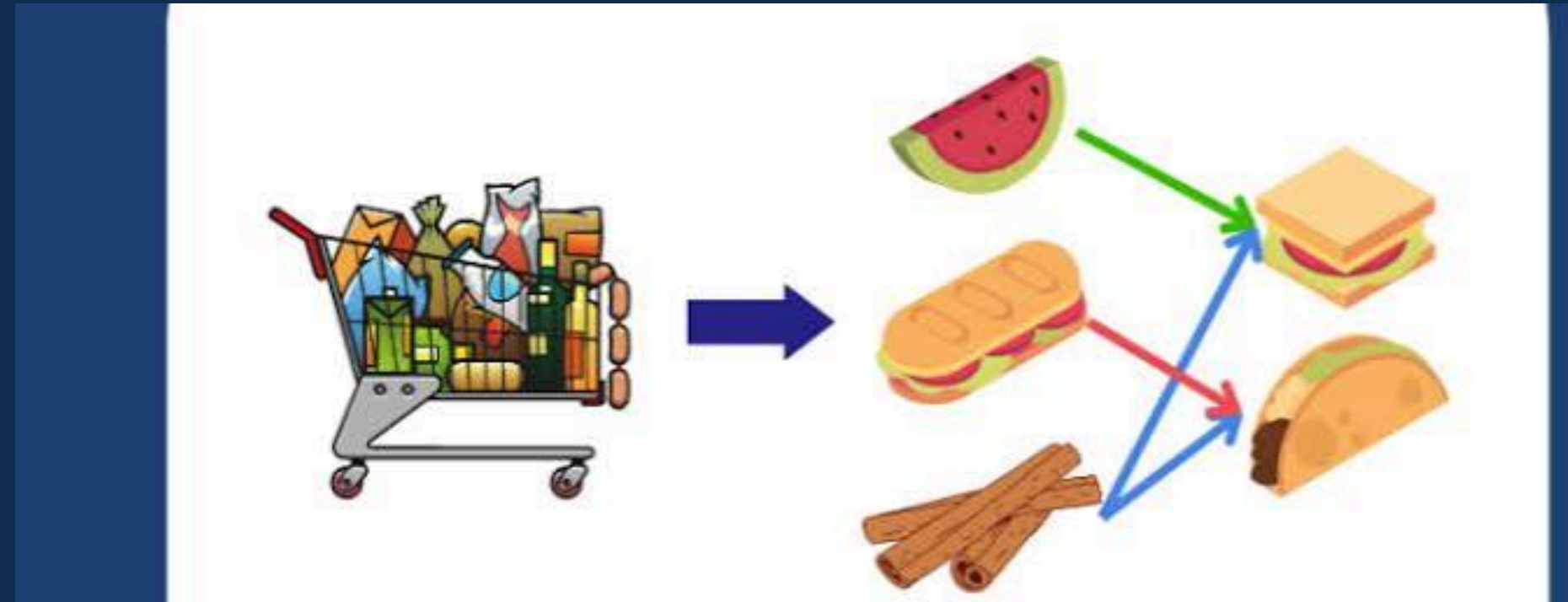
Example:

Bread $\rightarrow$ Butter

Where:

A = Antecedent

B = Consequent



Meaning

Customers who purchase A may also purchase B.

Important:

Association does not automatically mean causation.

Support

What is Support?

Support tells us:

How frequently an item or itemset appears
across all transactions.

$$Support(A) = \frac{\text{Transactions containing A}}{\text{Total Transactions}}$$

For an item pair :

$$Support(A, B) = \frac{\text{Transactions containing both A and B}}{\text{Total Transactions}}$$

Support

$$\textit{Support}(A, B) = \frac{\text{Transactions containing both A and B}}{\text{Total Transactions}}$$

Suppose there are:

10,000 transactions

And:

600 transactions contain Bread and Butter

$$\textit{Support}(A) = \frac{\text{Transactions containing A}}{\text{Total Transactions}}$$

Then:

$\text{Support}(\text{Bread}, \text{Butter}) = 600 / 10,000 = 0.06$

Therefore:

Support = 6%

Interpretation

6% of all transactions contain both Bread and Butter.

Confidence

What is Confidence?

Confidence tells us:

When A is purchased, how often is B also purchased?

$$\textit{Confidence}(A \rightarrow B) = \frac{\textit{Support}(A, B)}{\textit{Support}(A)}$$

OR

$$\textit{Confidence}(A \rightarrow B) = \frac{\text{Transactions containing A and B}}{\text{Transactions containing A}}$$

Confidence

$$Confidence(A \rightarrow B) = \frac{Support(A, B)}{Support(A)}$$

Confidence Example

Suppose:

1,000 customers purchased Bread

Among them:

600 also purchased Butter

Therefore:

$$Confidence(Bread \rightarrow Butter) = 600/1000 = 60\%$$

Interpretation

60% of Bread transactions also contained Butter.

$$Confidence(A \rightarrow B) = \frac{\text{Transactions containing A and B}}{\text{Transactions containing A}}$$

Lift

What is Lift?

Lift measures how much more frequently A and B occur together compared with what would be expected based on B's overall frequency.

Lift	General Meaning
> 1	Positive association
1	No additional association indicated
< 1	Negative association

$$Lift(A \rightarrow B) = \frac{Confidence(A \rightarrow B)}{Support(B)}$$

Suppose:

Confidence(Bread → Butter) = 60%

And:

Support(Butter) = 20%

Then:

Lift=0.60 / 0.20 =3

Interpretation

The occurrence of Butter among Bread transactions is 3 times its overall transaction rate.

SUPPORT vs CONFIDENCE vs LIFT

Measure	Main Question
Support	How common is this itemset?
Confidence	When A occurs, how often does B occur?
Lift	Is the A–B relationship stronger than the baseline expectation?

Easy way to remember :

Support → Popularity

Confidence → Reliability of the rule

Lift → Strength relative to baseline